St Mary's DSG, Kloof

Grade 10

10 May 2024

Mathematics Cycle Test

Time: 55 Minutes **Marks: 55**

NAME: _____Memo_____

MATHS TEACHER: _____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 8 pages and 8 questions.

2. Please check that your question paper is complete.

3. An approved, non-programmable calculator may be used.

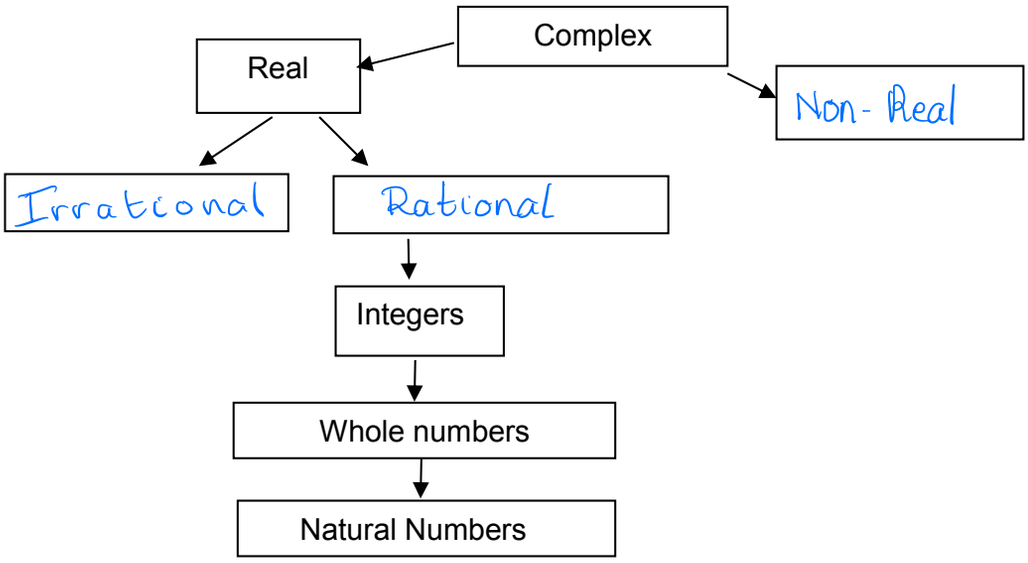
4. Answer questions on the question paper and show all working. Giving an answer only may not result in all the marks being awarded.

5. It is in your interest to write neatly.

6. Please show all working details.

Question 1

a) Complete the diagram representing the number system:



(3)

b) Consider the following numbers: $\sqrt{27}$; $\sqrt[3]{-27}$; $\sqrt{-27}$

Which ONE of these numbers is:

(1) Irrational? $\sqrt{27}$

(2) Non-real? $\sqrt{-27}$ (2)

c) If x is an integer, then will $x \div \sqrt{49}$ be rational or irrational? Rational (1)

d) For which value of x is $\frac{\sqrt{x-5}}{x+7}$ undefined? $x = -7$ (1)

e) 1. Convert $0,\dot{2}\dot{1}$ to a common fraction, **showing all working details**. (2)

Let $x = 0,\dot{2}\dot{1}$

$100x = 21,\dot{2}\dot{1}$

$99x = 21$

$x = \frac{21}{99}$

$x = \frac{7}{33}$

2. Express $7,\dot{2}\dot{1}$ as a mixed fraction. (2)

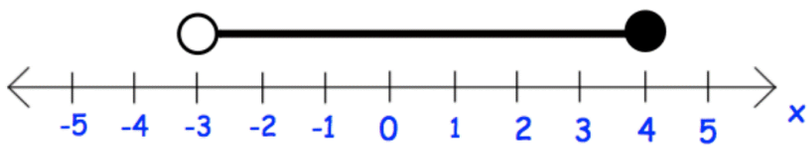
$7\frac{7}{33}$

[11]

Question 2

Multiple choice. Circle the correct option in each case.

a)



Which description illustrates the number line above.

1. $-3 < x < 4$
2. $-3 \leq x < 4$
3. $-3 < x \leq 4$
4. $-3 \leq x \leq 4$ (1)

b) The solution to the inequality $10 \geq x - 3$ is:

1. $x \in [7 ; \infty)$
2. $x \in (-\infty ; 7]$
3. $x \in (-\infty ; 13]$
4. $x \in [13 ; \infty)$ (1)

[2]

Question 3

Given the inequality: $-7 < 2x - 3 \leq 5$

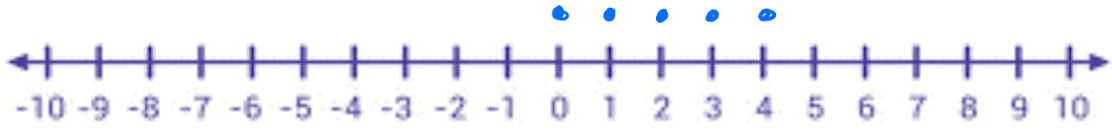
a) Solve the inequality for x. (2)

$-4 < 2x \leq 8$

$-2 < x \leq 4$

b) Represent the solution to the inequality for $x \in N_0$ on the number line below:

(2)



[4]

Question 4

Solve for the unknown:

a) $3x^2 + 10x - 8 = 0$ (2)

$(3x-2)(x+4) = 0$

$3x-2 = 0$ or $x+4 = 0$

$x = \frac{2}{3}$ $x = -4$

b) $(x-3)(x+2) = 14$ (4)

$x^2 + 2x - 3x - 6 - 14 = 0$

$x^2 - x - 20 = 0$

$(x-5)(x+4) = 0$

$x = 5$ or $x = -4$

c) Given $\frac{1}{pq} - \frac{1}{q} = \frac{1}{x}$

1) Solve for x. (4)

$\frac{x - px}{pqx} = \frac{pq}{pqx}$

$x(1-p) = pq$

$x = \frac{pq}{1-p}$

2) State any restrictions that apply. (2)

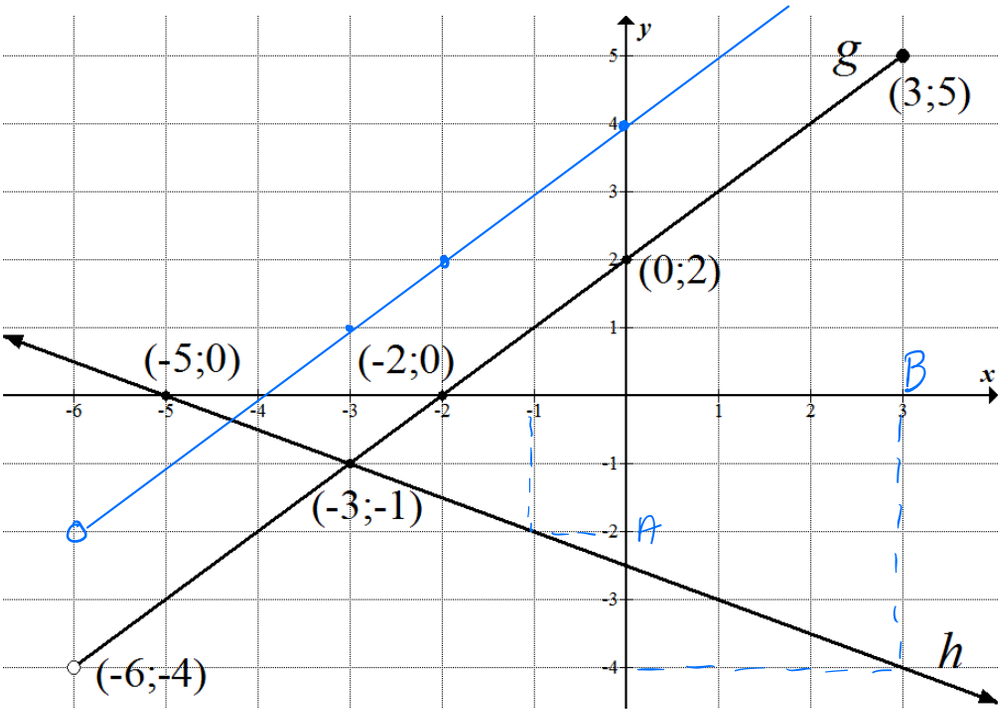
$p, q, x \neq 0$

$p \neq 1$

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Question 5

On the set of axes below the graphs of g and h have been drawn.



a) State the domain of **graph** g $x \in (-6;3]$ (2)

b) State the range of **graph** h $y \in \mathbb{R}$ (1)

c) Use the **graph of** h to answer the following questions: Show clearly **on the graph** using the letters A & B where your answers were read.

1. Find the value of y if $x = -1$ $y = -2$ (Read at A) (1)

2. Find the value of x if $y = -4$ $x = 3$ (Read at B) (1)

d) **Graph** g is shifted 2 units vertically to form a new **graph** f.
Draw graph f on the same system of axes. (2)

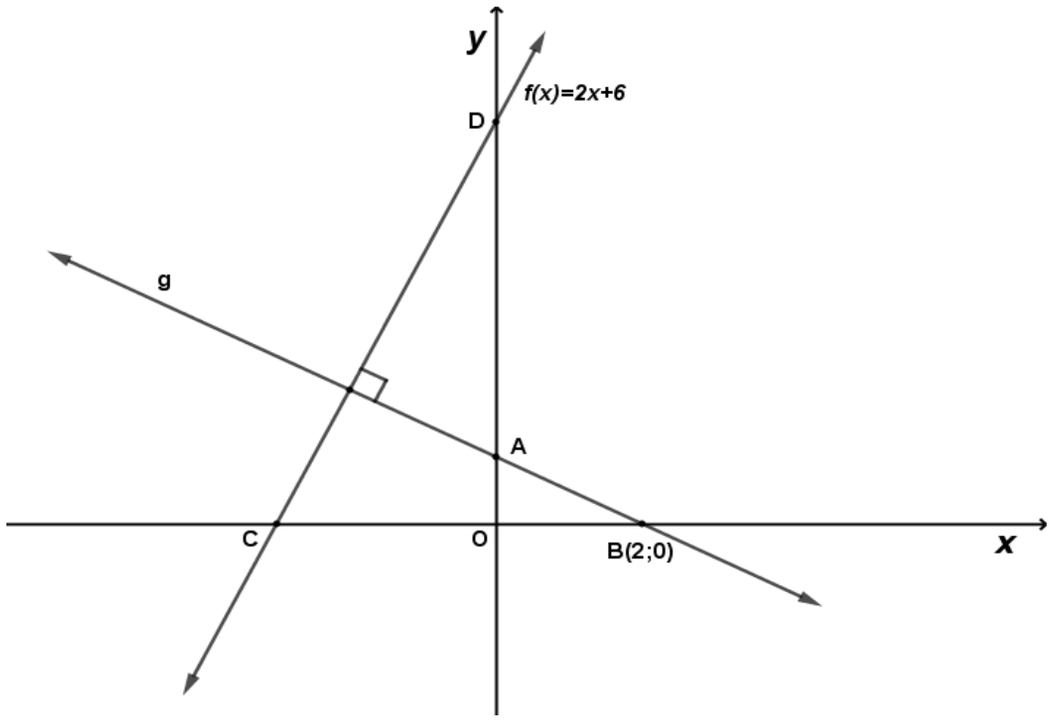
[7]

Question 6

Refer to the diagram below. Graph f is perpendicular to graph g

Graph $f(x) = 2x + 6$ intersects the x –axis at C and the y –axis at D.

Graph g intersects the x –axis at B (2;0) and the y-axis at A.



a) Determine the equation of *g*. $m = -\frac{1}{2}$ $(f \perp g)$ (3)

$y = -\frac{1}{2}x + c$

Sub (2;0) $0 = -\frac{1}{2}(2) + c$

$1 = c$

$y = -\frac{1}{2}x + 1$

b) Determine the equation of *h*, if *h* is drawn parallel to *f* and passes through B. (2)

$y = 2x + c$

Sub (2;0) $0 = 2(2) + c$

$-4 = c$

$y = 2x - 4$

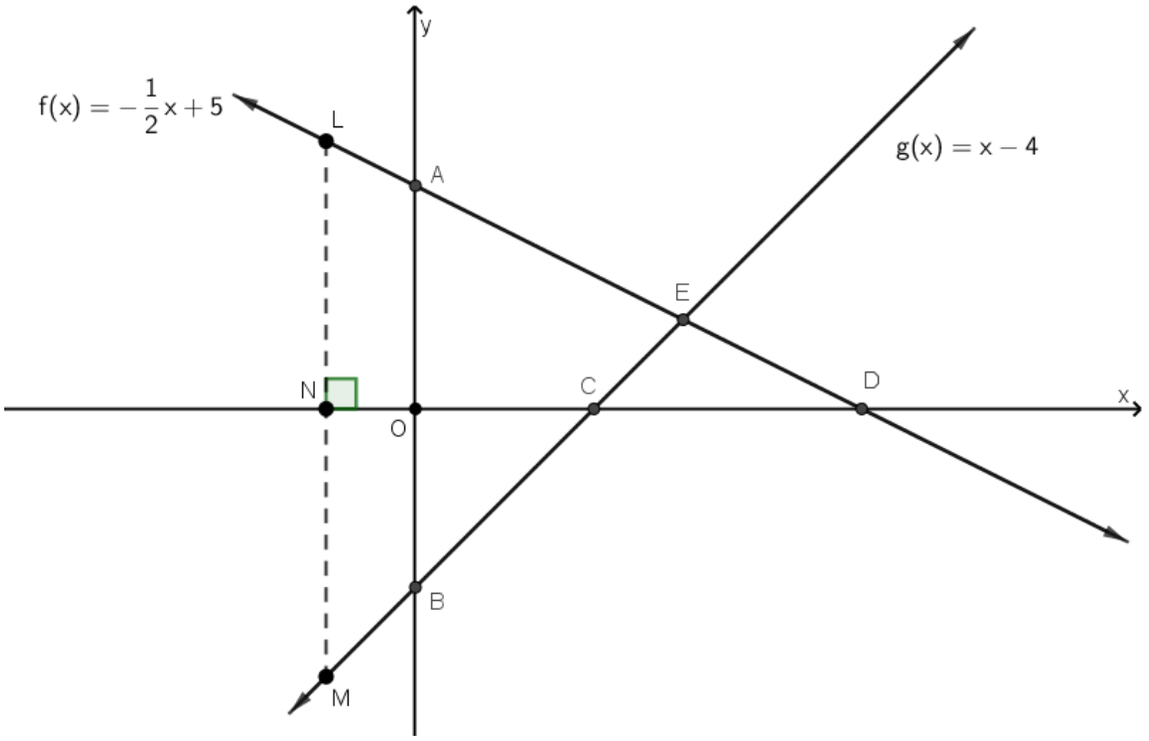
[5]

Question 7

Given functions $f(x) = -\frac{1}{2}x + 5$ and $g(x) = x - 4$.

E is the point of intersection of graphs f and g

LNM is parallel to the y-axis.



Determine:

a) the length of OB (1)

4 units

b) the length of OD (2)

$0 = -\frac{1}{2}x + 5$ D (x : 0)

$\frac{1}{2}x = 5$

$x = 10$ OD = 10 units

c) the length of the vertical line LM if it is given that ON = 2 units. (4)

M : $y = -2 - 4 = -6$ M (-2 : -6)

L : $y = -\frac{1}{2}(-2) + 5 = 6$ L (-2 : 6)

$\therefore LM = LN + NM$

$= 6 + 6$

$= 12$ units

d) Determine the coordinates of the point of intersection E of the two straight lines:

$f(x) = -\frac{1}{2}x + 5$ and $g(x) = x - 4$. (4)

$f(x) = g(x)$

$-\frac{1}{2}x + 5 = x - 4$

$9 = 1\frac{1}{2}x$

$6 = x$

Sub in $g(x)$

$y = 6 - 4 = 2$

$\therefore E(6;2)$

[11]

Question 8

In a group of 27 people, 17 of them have studied French, 7 have studied Chinese and 7 have not studied either. How many of them have studied both French and Chinese? Explain your reasoning.

$27 - 7 = 20$ people study French or Chinese

$17 + 7 = 24$

$\therefore$ 4 have studied both

[3]